1. Identity

Document	STAR-PKG-STAR-IOD-EN · v1.6 · 2026-10-21
Package	star-iod
Version	0.1.0
Capability	star.orbit.iod-gibbs
Maturity	RELEASED
Python	⩾3.9
Source commit	94fcf7c
Source	https://github.com/Andrea07072000/S.T.A.R.-/tree/main/packages/star_iod

2. Licence and obligations

Apache License 2.0. You may use, copy, modify and distribute it, also commercially, and you receive a patent licence from the contributors. When you redistribute it: keep the LICENSE file and the copyright notices (and a NOTICE file, if a future version adds one), and state in modified files that you changed them. You may not use the S.T.A.R. name to endorse your product.

3. What it needs at run time

Nothing beyond the Python standard library: it works offline.

4. Integrity

SHA-256 fingerprints of the files published on this website. Check a downloaded file with:

```
python -c "import hashlib,sys;print(hashlib.sha256(open(sys.argv[1],'rb').read()).hexdigest())" star_iod-0.1.0-py3-none-any.whl
```

FILE	SHA-256
star_iod-0.1.0-py3-none-any.whl	aa4cf7056e95a013a28d5a4270f713ee800f68cf6778bdc5e0563794adedc983
star_iod.py	6a2072071e695d5209723ff1f667502faff9e7f3c5683732a9765f1e4878d01e
star_iod-0.1.0-source.zip	bdd4a1594fc7591bab2b901ae8f59e56b4d9d580b2c4501af24b9498790b665b

5. What it does, and its limits

Velocity at the middle of three position vectors by Gibbs' method (cancellation-free form) and Herrick-Gibbs, with the measured domain of each (crossover near 0.4 degrees), agreement with Orekit IodGibbs within 8e-9 and explicit refusal of non-coplanar, collinear or impossible geometry.

Perfect positions in, velocity out: no measurement noise model, no covariance, no angles-only methods (Gauss, Laplace, Gooding), no differential correction. With noisy fixes the error of either method is dominated by the noise divided by the arc length, not by the figures above.

- `gibbs(r1, r2, r3, mu, coplanar_tol_deg=3)`: Vallado Algorithm 54. Reproduces Example 7-3 within 5e-7 km/s. The direction of motion is the order r1, r2, r3. The sums of the textbook form are written with differences of neighbouring vectors: at 0.02 degrees of spacing the error is 2e-8 relative instead of 1.5e-5.
- `herrick_gibbs(r1, r2, r3, t1, t2, t3, mu, coplanar_tol_deg=3)`: Vallado Algorithm 55, for closely spaced positions; `separation_deg` returns the two angles.
- Measured domains (240 known orbits, largest relative velocity error by spacing in mean anomaly):
- Every function returns three finite floats or raises `ValueError`: wrong shapes, non-numeric, boolean, NaN or infinite input; a zero position; mu <= 0; positions further than `coplanar_tol_deg` from a common plane; collinear or coincident positions; positions through which no orbit about the centre passes (Gibbs); times not strictly increasing (Herrick-Gibbs).

6. Export, use and certification

The package is published and freely available. That usually takes software out of export-control lists, but not every use of it: read the page on space activities and export control before using it in a controlled programme. It is not certified or qualified for flight; the evidence describes what was tested and how.

7. Citation and bill of materials

```
@software{cavazzini_star_iod_0_1_0,
  author  = {Cavazzini, Andrea},
  title   = {{star-iod}: Velocity from three positions (Gibbs, Herrick-Gibbs)},
  version = {0.1.0},
  year    = {2026},
  url     = {https://andrea07072000.github.io/S.T.A.R.-/software/star_iod/},
  note    = {Apache-2.0. Source: https://github.com/Andrea07072000/S.T.A.R.-, commit 94fcf7c}
}
```

CITATION.cff · BibTeX · SBOM CycloneDX 1.5 · SBOM SPDX 2.3

8. Warranty and liability

Provided "as is", without warranties or conditions of any kind (Apache License 2.0, sections 7 and 8), to the extent the law allows. This never excludes liability that the law does not allow to exclude.

Related: space activities and export control · acceptable use · evidence and limits · licence texts